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$$\int \frac{x^2 + x + 1}{x - 1} dx$$

$$\begin{array}{r|l} \begin{array}{rrr} x^2 & +x & +1 \\ -x^2 & +x & \\ \hline & 2x & +1 \\ & -2x & +2 \\ \hline & & 3 \end{array} & \begin{array}{r} x-1 \\ x+2 \\ \hline \end{array} \end{array}$$

$$= \int (x + 2) dx + \int \frac{3}{x - 1} dx$$

Substitutie: $u = x - 1 \Rightarrow du = dx$

$$= \int (x + 2) dx + 3 \int \frac{du}{u}$$

$$= \frac{x^2}{2} + 2x + 3 \cdot \ln |x - 1| + c$$

References